

Canonical Binomial Decomposition Does Not Preserve Real Roots

Two minimal counterexamples, exact cubic classifications,
and explicit failures in every degree

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Abstract

We give a negative answer to Question 3.10 of Mu and Welker as stated in arXiv:2503.24076v1. Their canonical recursive decomposition splits a polynomial with constant coefficient one as $F(t) = G(t) + tH(t)$ using the integer binomial expansion of each coefficient. Two cubic inputs are exhibited, and each is minimal in its own sense. The real-rooted polynomial $F(t) = (1+2t)(1+3t)^2$ has canonical outputs $G(t) = 1+7t+15t^2+8t^3$ and $H(t) = 1+6t+10t^2$, with discriminants -59 and -4 ; the simple-root polynomial $F(t) = (1+2t)(1+3t)(1+4t)$ has outputs $G(t) = 1+8t+19t^2+11t^3$ and $H(t) = 1+7t+13t^2$, with discriminants -31 and -3 . In both cases *both* outputs have nonreal zeros, and the second example removes repeated input roots as a possible explanation.

We prove that degree three is minimal. Exhaustive exact finite certificates show that 8 is the least linear coefficient among cubic counterexamples and 9 the least among cubic counterexamples with distinct input roots. For $F_m(t) = (1+mt)^3$ we determine all positive integers m for which either output is real-rooted: G_m precisely for $m \in \{1, 2, 6\}$, and H_m precisely for $m \in \{1, 2, 4, 6\}$. A separate family gives a complete classification for the second output when the input has three distinct roots.

The phenomenon is not confined to cubics, and two explicit thresholds are proved, neither implying the other. For every $d \geq 3$ and every integer $m \geq 20(d-1)$, the second output for the repeated-slope input $(1+mt)^d$ violates the first Newton inequality; for every $d \geq 3$ and every integer $n \geq 10d^3$, the same happens for the distinct-root consecutive-slope input $\prod_{j=0}^{d-1} (1+(n+j)t)$. We also derive a two-sided coefficientwise scaling limit, a general open-region dilation criterion, and the asymmetry that the first output eventually regains distinct real roots under dilation of a fixed simple-root input while the second never does. A compressed simplicial complex realizes the smaller example's full decomposition as a deletion and a link, while the same input also has a multipartite realization all of whose links are real-rooted: the failure is local, not a property of the f -vector. Full proofs, integer certificates, and two independent executable verification stacks are supplied. The Bell–Skandera simplicial-complex realization question and Mu–Welker Conjecture 3.8 are not resolved by these counterexamples.

Status and scope. This is an unrefereed research manuscript, not a proof-assistant formalization. The question and definitions were checked in the 31 March 2025 arXiv version, in the rendered PDF pages rather than only in search snippets or an experimental HTML conversion. A literature search on 20 September 2026 did not locate an earlier answer, but priority is not certified and the publisher's final full text was not inspected. The mathematical claims below refer to the explicitly defined decomposition, independently of that bibliographic limitation.

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1 The question and the outcome

Let

$$F(t) = 1 + c_1t + \cdots + c_d t^d, \quad c_k \in \mathbb{Z}_{>0}. \quad (1.1)$$

We call a nonzero real polynomial *real-rooted* if all its complex zeros are real, allowing repeated zeros. Constant nonzero polynomials are real-rooted by this convention. A polynomial of the form (1.1) cannot vanish at a nonnegative real number. Thus, if it is real-rooted, it factors as

$$F(t) = \prod_{i=1}^d (1 + \lambda_i t), \quad \lambda_i > 0. \quad (1.2)$$

The numbers λ_i need not be integers. Such a factorization certifies real-rootedness without any numerical root calculation.

Mu and Welker define a canonical coefficientwise splitting $F = G + tH$ in [1, Section 3.2]. Their Question 3.10 asks whether real-rootedness of F forces both outputs to be real-rooted. We answer this question negatively for that precise splitting. The word *recursive*, used for this decomposition in [1] and in part of the supplementary material, names this particular arithmetic rule; it does not mean an arbitrary choice of G and H with $F = G + tH$.

1.1 Two headline witnesses

Two cubic inputs carry the negative answer, and both are kept throughout because they are minimal in two different senses. The first has the smallest possible linear coefficient:

$$\begin{aligned} F(t) &= (1 + 2t)(1 + 3t)^2 = 1 + 8t + 21t^2 + 18t^3, \\ G(t) &= 1 + 7t + 15t^2 + 8t^3, \quad \text{Disc}(G) = -59, \\ H(t) &= 1 + 6t + 10t^2, \quad \text{Disc}(H) = -4. \end{aligned} \quad (1.3)$$

The second is the smallest with three *distinct* input roots, and so removes “repeated roots caused it” as an explanation:

$$\begin{aligned} F(t) &= (1 + 2t)(1 + 3t)(1 + 4t) = 1 + 9t + 26t^2 + 24t^3, \\ G(t) &= 1 + 8t + 19t^2 + 11t^3, \quad \text{Disc}(G) = -31, \\ H(t) &= 1 + 7t + 13t^2, \quad \text{Disc}(H) = -3. \end{aligned} \quad (1.4)$$

Each proof requires only three integer binomial expansions and two negative discriminants; neither depends on an extensive computation. They are proved in Theorems 3.1 and 3.3.

The extensions are substantial. We prove degree-minimality, certify both minimality statements *within cubics*, classify two infinite cubic families completely, and give two uniform constructions,

$$F(t) = (1+mt)^d \quad (d \geq 3, m \geq 20(d-1)), \quad F_{d,n}(t) = \prod_{j=0}^{d-1} (1+(n+j)t) \quad (d \geq 3, n \geq 10d^3), \quad (1.5)$$

for both of which H is not real-rooted. We then explain the large-coefficient mechanism by a limiting transform. In particular, multiple input roots are not essential, and neither threshold in (1.5) implies the other.

1.2 Results kept in pairs, and what each pair buys

This manuscript merges two independently written treatments of the same question (see Section 1.3). Several results come in pairs whose members look redundant but are not. Each pair is marked at its place in the text. We list them here with one sentence on what keeping both buys.

- P1. *Both cubic witnesses*, (1.3) and (1.4), each with its own digit table and discriminant pair (Theorems 3.1 and 3.3): the first is the smallest counterexample, and the second shows that a repeated input root is not what causes failure.
- P2. *Both minimality statements* (Propositions 4.3 and 4.4): 8 is least among cubic counterexamples while 9 is least among cubic counterexamples with distinct roots, so the two numbers answer two different questions and neither subsumes the other.
- P3. *Both all-degree families with their explicit thresholds* (Theorems 7.3 and 7.4): the repeated-slope family has the far smaller threshold $20(d-1)$, and the consecutive-slope family is the only one for which an explicit threshold is proved with *distinct* roots.
- P4. *Both scaling limits* in Theorem 8.2, on the H side and on the G side: only having both exhibits the asymmetry of Proposition 8.4, which is the sharpest qualitative statement about the first output.
- P5. *Both directions of the relation between the two failures* (Lemma 5.2 and the counterexample to its converse in (6.3)): the lemma alone would misstate the scope, since G can be real-rooted while H is not.
- P6. *Both simplicial realizations of the same input* $1 + 8t + 21t^2 + 18t^3$ (Sections 3.3 and 9): the multipartite join has only real-rooted links, the compressed complex has a non-real-rooted link, and together they prove that the f -vector does not determine the real-rootedness of every local link.
- P7. *Both decompositions of the repeated input*, the canonical one and the noncanonical one in (3.6): the second shows the negative answer is about the specified rule and not about the existence of some real-rooted splitting.
- P8. *Both verification stacks* (Section 10): one is independent inside Python, the other is independent of Python, and two different computer algebra systems confirming the same integers is worth more than either alone.
- P9. *Both standalone certificates* (Sections A and B): each is a hand-checkable disproof on its own, one driven from the search bounds and one from the literal digit table.
- P10. *Both cubic enumerations* (Tables 2 and 3): the wider scan is the stronger certificate, the narrower one is hand-auditable, and their agreement across two different search rectangles is itself a check.
- P11. *Both all-degree witness tables* (Table 6 and `results/all_degree_certificates.csv`): the dyadic table shows the proved thresholds are far from sharp, the certificate file audits a proved threshold exactly at and just above its stated value.
- P12. *Both provenance documents* (`REFERENCES.md` and `STATUS.md`): neither contains all of the page-level source audit, the search enumeration, the journal-metadata caveat and the indexing correspondence.

- P13. *Both rearrangements in the degree-two proof* (Theorem 4.1 and Remark 4.2): the second exhibits three separately nonnegative contributions, which the first hides.
- P14. *Both forms of the $k = 2$ estimate* (Lemma 2.3): the exact triangular-number form drives the transfer lemma, and the two-sided inequalities drive every asymptotic estimate; neither substitutes for the other.
- P15. *Both statements of what is not resolved* (Section 1.4): Bell–Skandera Question 1.1 and Mu–Welker Conjecture 3.8 are logically separate, and Proposition 3.9 of [1] is the bridge between them.

1.3 Provenance

This article is the merger of two research packages prepared on the same day for the same question, `mu-welker-binomial-decomposition` and `recursive-binomial-real-rootedness`. They proved the shared core theorem by the same route and differed only in which witness they headlined. The first package headlined the repeated-root input (1.3) and carried (1.4) as a proposition; the second headlined (1.4) and carried (1.3) in a subsection on smaller linear coefficients. Both computed the same three integer binomial expansions, the same two discriminant pairs, the same hockey-stick uniqueness argument, the same $k = 2$ and $k = 3$ estimates, the same first Newton inequality with the same Cauchy–Schwarz proof, the same leading-digit asymptotic, the same search rectangle for cubics, and the same conclusion that the Bell–Skandera question and Conjecture 3.8 are untouched. The present text proves that shared core once. Where the two genuinely diverged — the explicit all-degree thresholds — both results survive in full, in Theorems 7.3 and 7.4. The structure, notation (κ_k, R_k, F, G, H) and macros of the first package are used throughout; the second package’s $\mathcal{S}_k$, $\mathcal{G}_k$, f , g , h have been rewritten into them. The supplementary files of both packages are retained side by side, with the second package’s verifier renamed to `code/verify_wide.py`.

1.4 What is not resolved

The broader problem discussed in [1, Question 1.1] asks whether every real-rooted polynomial (1.1) is the face-counting polynomial of a simplicial complex. That problem, attributed there to Bell and Skandera, is logically different from Question 3.10, and we do not resolve it: our input examples already have explicit simplicial-complex realizations, so they cannot be obstructions to it. Nor do we disprove the coefficientwise inequality conjectured in [1, Conjecture 3.8]; it holds in every example exhibited below, as recorded explicitly in Sections 3.3 and 9. Proposition 3.9 of [1] is the bridge between those two statements, relating the coefficientwise conjecture to the realization question; nothing here bears on that bridge either. These are two independent non-results and both are stated deliberately.

2 The canonical binomial splitting

2.1 Integer binomial expansions

For integers $a \geq 0$ and $k \geq 0$, we use $\binom{a}{k} = 0$ when $a < k$, and $\binom{0}{0} = 1$. Negative upper indices are never needed. For each positive integer q and each positive integer k , there is a unique representation

$$q = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \cdots + \binom{a_j}{j}, \quad a_k > a_{k-1} > \cdots > a_j \geq j \geq 1. \quad (2.1)$$

The lower indices are consecutive; the expansion stops when the remainder becomes zero. We assign zero the empty expansion. Each polynomial coefficient receives its own expansion: the upper indices in the expansion of c_k are not shared with those in the expansion of c_{k+1} .

For completeness, the greedy construction is elementary. Choose the largest a_k such that $\binom{a_k}{k} \leq q$. If r is the remainder, then

$$0 \leq r < \binom{a_k+1}{k} - \binom{a_k}{k} = \binom{a_k}{k-1}.$$

If $r > 0$, the next upper index is therefore less than a_k . Continue with lower index $k-1$. At lower index one, any positive remainder is represented by one binomial coefficient, so the procedure terminates.

To see uniqueness, fix the first upper index a . Every permissible sum of lower terms is at most

$$\sum_{\ell=1}^{k-1} \binom{a-k+\ell}{\ell} = \binom{a}{k-1} - 1,$$

by the elementary hockey-stick identity. Consequently any representation with first term $\binom{a}{k}$ lies in $[(\binom{a}{k}, \binom{a+1}{k})$. The first term is forced by q , and induction forces every remaining term. This also proves that the integer algorithm used in the supplementary code implements the specified expansion, and it is why an exact binary search for the leading index is sufficient.

2.2 Conventions

All polynomial coefficient lists in the supplementary JSON and CSV files are in *ascending* power order. A decomposition list may retain zero trailing coefficients; those are removed before classifying a polynomial's degree, so a trailing zero is never mistaken for an extra zero of the polynomial. The source [1] writes f_{i-1} for the coefficient of t^i ; we write c_i for that coefficient and keep the corresponding indexing in both outputs.

2.3 The splitting operator

Definition 2.1. For the expansion (2.1), define

$$\kappa_k(q) = \sum_{\ell=j}^k \binom{a_\ell-1}{\ell-1}, \quad R_k(q) = \sum_{\ell=j}^k \binom{a_\ell-1}{\ell}. \quad (2.2)$$

Set $\kappa_k(0) = R_k(0) = 0$. For (1.1), set

$$G_F(t) = 1 + \sum_{k=1}^d R_k(c_k)t^k, \quad H_F(t) = \sum_{k=1}^d \kappa_k(c_k)t^{k-1}. \quad (2.3)$$

Pascal's identity gives

$$R_k(q) + \kappa_k(q) = q, \quad F(t) = G_F(t) + tH_F(t). \quad (2.4)$$

For $q > 0$, $\kappa_1(q) = 1$. Hence both output polynomials have constant coefficient one. All their coefficients are nonnegative, and H_F has positive coefficients and degree exactly $d - 1$. The polynomial G_F may have degree smaller than d .

These are the outputs denoted g and h in [1, Section 3.2]. In particular, H_F is *not* the usual h -polynomial of a simplicial complex.

Remark 2.2 (A warning about other binomial operators). The upper index in both sums in (2.2) is lowered by one. Replacing $\binom{a_{\ell-1}}{\ell-1}$ by $\binom{a_{\ell}}{\ell-1}$ would give a different operator and would not be the question under consideration. The operation is also not the one that decrements only the *lower* indices, and it is not obtained by differentiating the polynomial. Concretely, for the simple-root example (1.4),

$$\kappa_3(24) = 13, \quad \text{whereas} \quad \frac{1}{3}[t^2]F'(t) = 24.$$

Confusing these constructions changes the question. The digit tables in Section 3 make the choice of operation explicit, summand by summand.

2.4 Two elementary estimates

We isolate bounds that will be used repeatedly. Lemma 2.3 is kept in both of its forms; this is pair P14.. The exact form drives the root-location argument of Lemma 5.2, where an inequality would not suffice; the two-sided inequalities drive every asymptotic estimate in Sections 7 and 8, where the exact form is unusable.

Lemma 2.3. *For every integer $q \geq 1$,*

$$\kappa_2(q) = \left\lceil \frac{\sqrt{8q+1} - 1}{2} \right\rceil. \quad (2.5)$$

Equivalently, $\kappa_2(q)$ is the smallest positive integer u such that $q \leq u(u+1)/2$; writing $q = \binom{n}{2} + r$ with $0 \leq r < n$, it equals $n - 1 + \mathbf{1}_{r>0}$. In particular,

$$\sqrt{q} \leq \kappa_2(q) < \sqrt{2q} + 1. \quad (2.6)$$

Proof. Write $q = \binom{n}{2} + r$, where $0 \leq r < n$. If $r = 0$, then $\kappa_2(q) = n - 1$. If $r > 0$, the remaining term is $\binom{r}{1}$, and $\kappa_2(q) = (n - 1) + 1 = n$. These are exactly the successive triangular-number intervals described in the statement.

If $u = \kappa_2(q)$, then $q \leq u(u+1)/2 \leq u^2$, which proves the lower bound. For the upper bound, put $x = (\sqrt{8q+1} - 1)/2$. Since $x < \sqrt{2q}$ and $\lceil x \rceil < x + 1$, formula (2.5) gives the result. $\square$

Lemma 2.4. *For every integer $q \geq 1$, with $x = (6q)^{1/3}$,*

$$\kappa_3(q) > \frac{(x-1)(x-2)}{2} = \frac{x^2}{2} - \frac{3x}{2} + 1. \quad (2.7)$$

Proof. Let n be maximal with $\binom{n}{3} \leq q$. Then

$$q < \binom{n+1}{3} = \frac{n^3 - n}{6} < \frac{n^3}{6},$$

so $n > x$. The first term in the expansion contributes $\binom{n-1}{2}$ to $\kappa_3(q)$; all other terms contribute nonnegative integers. The function $(y-1)(y-2)/2$ is strictly increasing for $y > 3/2$, and $x \geq 6^{1/3} > 3/2$. Hence

$$\kappa_3(q) \geq \binom{n-1}{2} > \frac{(x-1)(x-2)}{2}. \quad \square$$

3 An exact negative answer

The two theorems of this section are pair P1.: the first gives the smallest counterexample there is, and the second removes the possibility that a repeated input root is what causes the failure. Neither is a corollary of the other, and neither is demoted to a remark.

3.1 The minimal witness

Theorem 3.1. *The polynomial $F(t) = (1+2t)(1+3t)^2$ satisfies every input hypothesis of Mu–Welker Question 3.10 as stated in arXiv:2503.24076v1, but neither output in its canonical decomposition is real-rooted.*

Proof. The input has positive integer coefficients, constant coefficient one, and zeros $-1/2$ and $-1/3$, the latter of multiplicity two. Its relevant canonical expansions are

$$8 = \binom{8}{1}, \quad 21 = \binom{7}{2}, \quad 18 = \binom{5}{3} + \binom{4}{2} + \binom{2}{1}. \quad (3.1)$$

The upper indices $5 > 4 > 2$ in the last expansion satisfy the strict ordering requirement. Applying (2.2) gives the following exact table.

k	c_k	$R_k(c_k)$	$\kappa_k(c_k)$
1	8	7	1
2	21	$\binom{6}{2} = 15$	$\binom{6}{1} = 6$
3	18	$\binom{4}{3} + \binom{3}{2} + \binom{1}{1} = 8$	$\binom{4}{2} + \binom{3}{1} + \binom{1}{0} = 10$

Thus the outputs are precisely those in (1.3). The quadratic H has discriminant $6^2 - 4 \cdot 10 = -4$, and its zeros are

$$\frac{-3+i}{10}, \quad \frac{-3-i}{10}. \quad (3.2)$$

For a cubic $1 + at + bt^2 + ct^3$, write

$$D(a, b, c) = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc. \quad (3.3)$$

This is its discriminant. Direct substitution gives

$$D(7, 15, 8) = 11025 - 13500 - 10976 - 1728 + 15120 = -59.$$

A real cubic with negative discriminant has one real zero and one nonreal conjugate pair. Hence G also fails to be real-rooted. $\square$

Remark 3.2 (Why the discriminant test is exact). For a real cubic with leading coefficient $c \neq 0$ and roots r_1, r_2, r_3 , its discriminant is $c^4 \prod_{i < j} (r_i - r_j)^2$. Three distinct real roots give positive discriminant. A real root together with $u \pm iv$, $v \neq 0$, gives a negative factor $(2iv)^2$ times a positive real factor. A zero discriminant means repeated roots; a repeated nonreal root and its conjugate would require degree at least four. Thus a real cubic is real-rooted exactly when its discriminant is nonnegative. No floating-point approximation is involved.

3.2 The minimal witness with distinct input roots

Theorem 3.3. *The simple-root polynomial*

$$F(t) = (1 + 2t)(1 + 3t)(1 + 4t) = 1 + 9t + 26t^2 + 24t^3$$

has three distinct negative zeros, and neither canonical output is real-rooted.

Proof. The input zeros are $-1/2, -1/3, -1/4$, which are distinct. The complete digit table is Table 1: it displays the two contributions summand by summand, so the action of Definition 2.1 can be followed term by term. The coefficient identity is immediate,

$$(1 + 8t + 19t^2 + 11t^3) + t(1 + 7t + 13t^2) = 1 + 9t + 26t^2 + 24t^3.$$

The quadratic output has discriminant $7^2 - 4 \cdot 13 = -3$ and zeros

$$\frac{-7 + i\sqrt{3}}{26}, \quad \frac{-7 - i\sqrt{3}}{26}, \quad (3.4)$$

which alone disproves the assertion that both outputs must be real-rooted. For the cubic output, (3.3) gives

$$D(8, 19, 11) = 19^2 \cdot 8^2 - 4 \cdot 11 \cdot 8^3 - 4 \cdot 19^3 - 27 \cdot 11^2 + 18 \cdot 11 \cdot 19 \cdot 8 = -31.$$

By Remark 3.2 it has one real zero and a nonreal conjugate pair. □

k	Canonical expansion of c_k	$R_k(c_k)$	$\kappa_k(c_k)$
1	$9 = \binom{9}{1}$	8	1
2	$26 = \binom{7}{2} + \binom{5}{1}$	$15 + 4 = 19$	$6 + 1 = 7$
3	$24 = \binom{6}{3} + \binom{3}{2} + \binom{1}{1}$	$10 + 1 + 0 = 11$	$10 + 2 + 1 = 13$

Table 1: The complete arithmetic certificate for the simple-root witness (1.4). Every entry is small enough to check by hand.

3.3 What the examples do not contradict

A simplicial complex on a finite vertex set is a family of subsets closed under taking subsets; its face polynomial is $\sum_{\sigma} t^{|\sigma|}$, including the empty face. Partition a vertex set into blocks of sizes $a_1, \dots, a_d$,

and allow faces to contain at most one vertex from each block. Independent choices in the blocks give the face polynomial

$$\prod_{i=1}^d (1 + a_i t). \quad (3.5)$$

Thus the main example, with block sizes 2, 3, 3, already has a realization, as does (1.4) with block sizes 2, 3, 4. This construction also applies to all integer-slope products used below, so every input in this manuscript is an f -polynomial and none of them can obstruct [1, Question 1.1]. This realization is one half of pair P6.; the other half, a differently shaped complex with the same face counts, is constructed in Section 9.

Even the two outputs in (1.3) are face polynomials. For H , take the complete graph on five vertices together with a sixth isolated vertex, without two-dimensional faces. For G , take all fifteen edges on six vertices, add a seventh isolated vertex, and select any eight of the ten triangular faces on the first five vertices. These are valid complexes with the prescribed face counts. The first complex is a subcomplex of the second. These two realizations serve a different purpose from those of Section 9: they show that the *outputs* are themselves f -polynomials, so no realizability obstruction is being exhibited on either side of the splitting.

Their coefficient sequences also satisfy

$$(1, 6, 10, 0) \leq (1, 7, 15, 8), \quad (1, 7, 13, 0) \leq (1, 8, 19, 11)$$

coordinatewise, after padding the shorter vector with a trailing zero. So neither example violates the separate inequality in Mu–Welker Conjecture 3.8. Failure of real-rootedness of G or H is logically different from failure of these inequalities.

There is also a different, noncanonical split with real-rooted outputs:

$$(1 + 2t)(1 + 3t)^2 = \underbrace{(1 + t)(1 + 3t)^2}_{1+7t+15t^2+9t^3} + t \underbrace{(1 + 3t)^2}_{1+6t+9t^2}. \quad (3.6)$$

Comparing this with (1.3), the canonical rule transfers just one unit from the cubic coefficient of the first output to the quadratic coefficient of the second. Real-rootedness is lost in both. This is pair P7.: without (3.6) a reader could not see that the negative answer concerns the specified canonical rule and not the existence of some real-rooted decomposition.

4 Minimality

4.1 No counterexample in degree at most two

Theorem 4.1. *For every real-rooted polynomial of the form (1.1) of degree one or two, both canonical outputs are real-rooted. Therefore three is the minimum possible degree of a counterexample.*

Proof. For $F = 1 + at$, the outputs are $G = 1 + (a - 1)t$ and $H = 1$.

Now let $F = 1 + at + bt^2$, with a, b positive integers. Real-rootedness means $a^2 \geq 4b$, hence $a \geq 2\sqrt{b}$. Put $u = \kappa_2(b)$. The outputs are

$$G = 1 + (a - 1)t + (b - u)t^2, \quad H = 1 + ut.$$

The second is linear. The first has nonnegative leading coefficient $b - u = R_2(b)$. If this coefficient is zero, G has degree at most one. Otherwise, because $2\sqrt{b} - 1 \geq 1$, its discriminant satisfies

$$\begin{aligned} (a-1)^2 - 4(b-u) &\geq (2\sqrt{b}-1)^2 - 4b + 4u \\ &= 1 + 4(u - \sqrt{b}) \geq 1, \end{aligned}$$

where the last inequality uses Lemma 2.3. Thus G is real-rooted. The example in Theorem 3.1 completes the degree-minimality claim. $\square$

Remark 4.2 (The same estimate, factored). This is pair P13.. The same discriminant can be organized as

$$(a-1)^2 - 4(b-u) = (a-2\sqrt{b})(a+2\sqrt{b}-2) + 4(u-\sqrt{b}) + 1 \geq 1, \quad (4.1)$$

which exhibits three separately nonnegative contributions: the first factor pair is nonnegative because $a \geq 2\sqrt{b}$ and $b \geq 1$, the middle term is nonnegative by Lemma 2.3, and the last is the constant 1. The rearrangement in the proof above hides this decomposition, so both are recorded.

4.2 The smallest linear coefficients among cubics

The following two statements are finite, exact computer-assisted refinements. Their search bounds are proved here, and the entire enumeration is supplied as CSV and JSON certificates and as executable code. They are pair P2.: the first answers “how small can a cubic counterexample be?”, the second answers “how small can one with distinct input roots be?”, and the two answers are different integers with different witnesses.

Proposition 4.3. *Among real-rooted cubics $1 + at + bt^2 + ct^3$ with positive integer coefficients, there is no counterexample with $a \leq 7$. With $a = 8$, the unique counterexample is $(a, b, c) = (8, 21, 18)$, the input of (1.3).*

Proof. Write the input as $\prod_{i=1}^3 (1 + \lambda_i t)$ with $\lambda_i > 0$. Then

$$a = \lambda_1 + \lambda_2 + \lambda_3, \quad b = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3, \quad c = \lambda_1\lambda_2\lambda_3.$$

Since $c \geq 1$, arithmetic–geometric mean gives $a \geq 3$ and $b \geq 3c^{2/3} \geq 3$. Also $b \leq a^2/3$ and $c \leq a^3/27$. Thus the complete finite region for $a \leq 8$ is

$$3 \leq a \leq 8, \quad 3 \leq b \leq \lfloor a^2/3 \rfloor, \quad 1 \leq c \leq \lfloor a^3/27 \rfloor. \quad (4.2)$$

For each triple, compute $D(a, b, c)$ from (3.3). By Remark 3.2, retaining exactly those with $D \geq 0$ selects exactly the real-rooted inputs in this region. Compute their canonical outputs by the greedy rule and test each output by its actual degree and discriminant. This gives Table 2. The sole failed output pair is (1.3). The complete retained list has 124 rows in `data/cubics_a_1e8.csv`; `verify.py` regenerates it directly. $\square$

Proposition 4.4. *Among real-rooted cubics $1 + at + bt^2 + ct^3$ with positive integer coefficients and distinct zeros, there is no counterexample with $a \leq 8$, and with $a = 9$ the input of (1.4), namely $(a, b, c) = (9, 26, 24)$, is a counterexample. Hence 9 is the least linear coefficient among cubic counterexamples with distinct input roots, while 8 is the least without that restriction.*

a	Candidate triples	Real-rooted inputs	Counterexamples
3	1	1	0
4	6	2	0
5	24	6	0
6	80	16	0
7	168	33	0
8	342	66	1
Total	621	124	1

Table 2: Complete exact search in (4.2), using the arithmetic–geometric lower bound $b \geq 3$. A counterexample means that at least one canonical output is not real-rooted. This is the hand-auditable half of pair P10..

Proof. Relax the lower bound on b to $b \geq 1$ and extend the scan to $a \leq 14$, which is legitimate because (4.2) bounds b and c from above for every a and the additional triples with $b < 3$ are merely never real-rooted. Table 3 records the resulting exhaustive enumeration. By Proposition 4.3 there is no counterexample at all for $a \leq 7$; the unique one at $a = 8$ has $D(8, 21, 18) = 0$, hence a repeated zero by Remark 3.2. At $a = 9$ the triple $(9, 26, 24)$ has $D = 4 > 0$, so its zeros are distinct, and Theorem 3.3 shows both outputs fail. $\square$

a	Candidate (b, c) pairs	Real-rooted inputs	Failures
1–2	0	0	0
3	3	1	0
4	10	2	0
5	32	6	0
6	96	16	0
7	192	33	0
8	378	66	1
9	729	120	2
10	1221	203	1
11	1960	329	0
12	3072	512	6
13	4536	762	8
14	6565	1109	7
Total	18794	3159	25

Table 3: The wider exact scan, using the weaker lower bound $b \geq 1$. A “failure” means that at least one output is not real-rooted; the two outputs need not both fail. These are not counts over all degrees. This is the stronger half of pair P10.; the complete failure list is in `results/cubic_failures.json`.

Remark 4.5 (Why both scans are kept). This is pair P10.. Table 2 uses the rectangle $3 \leq b \leq \lfloor a^2/3 \rfloor$, Table 3 the rectangle $1 \leq b \leq \lfloor a^2/3 \rfloor$; the candidate counts therefore differ (342 versus 378 at $a = 8$, for instance) while the real-rooted counts agree exactly, row by row, everywhere the two overlap. That agreement across two differently specified search regions, produced by two programs that share no code, is itself a check. Table 2 is small enough to audit by hand against the 124-row CSV;

Table 3 is the stronger certificate and is the only place where the vanishing of the failure count at $a = 11$, between two nonzero rows, is visible.

The restriction to cubics is essential to the wording of both Propositions 4.3 and 4.4. We have not proved that eight is the smallest linear coefficient among counterexamples of *all* degrees, nor that nine is smallest among simple-root counterexamples of all degrees. The unconditional degree-minimality result is Theorem 4.1.

5 A complete classification for repeated-root cubics

For a positive integer m , write

$$\begin{aligned} F_m(t) &= (1 + mt)^3 = 1 + 3mt + 3m^2t^2 + m^3t^3, \\ u_m &= \kappa_2(3m^2), \quad v_m = \kappa_3(m^3), \\ H_m(t) &= 1 + u_mt + v_mt^2, \\ G_m(t) &= 1 + (3m - 1)t + (3m^2 - u_m)t^2 + (m^3 - v_m)t^3. \end{aligned} \tag{5.1}$$

Theorem 5.1. *For positive integers m ,*

$$H_m \text{ is real-rooted} \iff m \in \{1, 2, 4, 6\}, \tag{5.2}$$

$$G_m \text{ is real-rooted} \iff m \in \{1, 2, 6\}. \tag{5.3}$$

In particular, both outputs have a nonreal conjugate pair for every $m \geq 7$.

5.1 An analytic tail for the quadratic output

Put $\rho = 6^{1/3}$. From Lemmas 2.3 and 2.4,

$$u_m < \sqrt{6}m + 1, \quad v_m > \frac{\rho^2 m^2}{2} - \frac{3\rho m}{2} + 1.$$

It follows that the discriminant $\delta_m = u_m^2 - 4v_m$ satisfies

$$\delta_m < (6 - 2\rho^2)m^2 + (6\rho + 2\sqrt{6})m - 3. \tag{5.4}$$

The inequalities

$$\rho^2 > \frac{33}{10}, \quad \rho < \frac{11}{6}, \quad \sqrt{6} < \frac{5}{2}$$

are certified by cubing or squaring positive quantities: $(33/10)^3 < 36$, $(11/6)^3 > 6$, and $(5/2)^2 > 6$. Therefore

$$\delta_m < -\frac{3}{5}m^2 + 16m - 3 < 0 \quad (m \geq 27). \tag{5.5}$$

The final quadratic is negative at $m = 27$ and strictly decreasing from that point on. This is a proof for the entire infinite tail, not an extrapolation from a table.

For $1 \leq m \leq 26$, direct integer binomial expansions give Table 4. The entries with nonnegative δ_m occur exactly at $m = 1, 2, 4, 6$. Together with (5.5), this proves (5.2).

m	u_m	v_m	$\text{Disc}(H_m)$	$D(3m-1, 3m^2-u_m, m^3-v_m)$
1	2	1	0	0
2	5	6	1	5
3	7	14	-7	-147
4	10	25	0	-31
5	12	39	-12	-556
6	15	55	5	32
7	17	77	-19	-1612
8	20	102	-8	-1156
9	22	130	-36	-5279
10	24	161	-68	-13231
11	27	191	-35	-6960
12	29	229	-75	-18700
13	32	272	-64	-18775
14	34	317	-112	-40207
15	37	365	-91	-36204
16	39	414	-135	-61691
17	42	464	-92	-44727
18	44	523	-156	-87703
19	47	586	-135	-84699
20	49	651	-203	-144896
21	51	719	-275	-225644
22	54	780	-204	-165836
23	56	857	-292	-273240
24	59	937	-267	-268192
25	61	1020	-359	-407595
26	64	1105	-324	-388287

Table 4: Exact finite part of the repeated-cubic classification. At $m = 1$, $G_m = (1+t)^2$ has degree two; the last column is the cubic *formula* with a zero cubic coefficient, not a claim of degree three. All other G_m in the table have degree three.

5.2 Why failure transfers to the cubic output

A root-location argument is more informative than estimating a second large discriminant.

Lemma 5.2. *For $m \geq 2$, if H_m is not real-rooted, then G_m is not real-rooted.*

Proof. The leading coefficient v_m of H_m is positive. Thus a negative quadratic discriminant gives $H_m(t) > 0$ for every real t . For $-1/m \leq t < 0$, the identity

$$G_m(t) = (1 + mt)^3 - tH_m(t)$$

then implies $G_m(t) > 0$. For $t \geq 0$, nonnegative coefficients and constant coefficient one also imply $G_m(t) > 0$.

For $m \geq 2$, G_m genuinely has degree three. Indeed, $m^3 \geq 8$ makes the first upper index n in the third binomial expansion at least four, so $R_3(m^3) \geq \binom{n-1}{3} > 0$.

Suppose, for contradiction, that all three roots of G_m are real. They must all lie strictly to the left of $-1/m$, so

$$G_m(t) = \prod_{i=1}^3 (1 + \lambda_i t), \quad 0 < \lambda_i < m.$$

Let $\varepsilon_i = m - \lambda_i > 0$. Comparing linear coefficients gives $\sum_i \varepsilon_i = 1$. Comparing quadratic coefficients now gives

$$3m^2 - u_m = 3m^2 - 2m + \sum_{i < j} \varepsilon_i \varepsilon_j,$$

so $u_m = 2m - \sum_{i < j} \varepsilon_i \varepsilon_j < 2m$. But

$$3m^2 > \frac{(2m)(2m+1)}{2} \quad (m > 1),$$

and the triangular-number characterization in Lemma 2.3 implies $u_m > 2m$. This contradiction proves the lemma. $\square$

Completion of Theorem 5.1. The classification of H_m and Lemma 5.2 exclude every m except 1, 2, 4, 6 for G_m . At $m = 4$, its discriminant is -31 , so it also fails: here H_4 succeeds while G_4 fails. The other three cases can be verified without a cubic root formula:

$$\begin{aligned} G_1(t) &= (1 + t)^2, \\ G_2(t) &= (1 + 2t)(1 + 3t + t^2), \\ G_6(t) &= (1 + 7t)(1 + 10t + 23t^2). \end{aligned}$$

The two quadratic factors have positive discriminants 5 and 8. This proves (5.3) and the assertion about $m \geq 7$. $\square$

The isolated successful values $m = 4$ for H_m and $m = 6$ for both outputs show that the small-parameter behavior is not monotone. An infinite-tail proof is therefore important even when many finite values have been tested.

6 A complete quadratic-output classification with simple input roots

Consider

$$\tilde{F}_m(t) = (1 + 2mt)(1 + 3mt)(1 + 4mt) = 1 + 9mt + 26m^2t^2 + 24m^3t^3. \quad (6.1)$$

All three input roots are distinct for every positive integer m . Put

$$\tilde{u}_m = \kappa_2(26m^2), \quad \tilde{v}_m = \kappa_3(24m^3).$$

Its quadratic output is $\tilde{H}_m = 1 + \tilde{u}_mt + \tilde{v}_mt^2$.

Theorem 6.1. *The polynomial $\tilde{H}_m$ is real-rooted exactly when $m = 3$ or $m = 4$. Thus it has a nonreal conjugate pair for $m = 1, 2$ and for every $m \geq 5$.*

Proof. Let $\rho = 144^{1/3}$. By Lemmas 2.3 and 2.4,

$$\tilde{u}_m < \sqrt{52}m + 1, \quad \tilde{v}_m > \frac{\rho^2 m^2}{2} - \frac{3\rho m}{2} + 1.$$

Hence

$$\text{Disc}(\tilde{H}_m) < (52 - 2\rho^2)m^2 + (6\rho + 2\sqrt{52})m - 3.$$

The rational bounds

$$\rho^2 > \frac{137}{5}, \quad \rho < \frac{21}{4}, \quad \sqrt{52} < \frac{29}{4}$$

follow from $(137/5)^3 < 144^2$, $(21/4)^3 > 144$, and $(29/4)^2 > 52$. Thus

$$\text{Disc}(\tilde{H}_m) < -\frac{14}{5}m^2 + 46m - 3 < 0 \quad (m \geq 17). \quad (6.2)$$

Again the last quadratic is negative at the stated endpoint and decreases thereafter. The finite computations in Table 5 cover all remaining values. Only $m = 3, 4$ give nonnegative discriminants. $\square$

6.1 No converse to the transfer lemma

There is no general converse to Lemma 5.2 outside its particular repeated-root family. Already at $m = 2$ in (6.1),

$$\begin{aligned} \tilde{G}_2 &= 1 + 17t + 90t^2 + 140t^3, & \text{Disc}(\tilde{G}_2) &= 20 > 0, \\ \tilde{H}_2 &= 1 + 14t + 52t^2, & \text{Disc}(\tilde{H}_2) &= -12 < 0. \end{aligned} \quad (6.3)$$

So the first output can be real-rooted while the second is not. Together with Lemma 5.2 this is pair P5.: keeping only the lemma would misstate the scope of the relation between the two failures, since it holds in exactly one direction and only for the repeated-slope cubics. In Section 8 we prove that $\tilde{G}_m$ is real-rooted for all sufficiently large m , even though $\tilde{H}_m$ is not. No exact classification of $\tilde{G}_m$ for all m is asserted here.

m	$\tilde{u}_m$	$\tilde{v}_m$	$\text{Disc}(\tilde{H}_m)$
1	7	13	-3
2	14	52	-12
3	22	118	12
4	29	210	1
5	36	337	-52
6	43	486	-95
7	50	660	-140
8	58	860	-76
9	65	1101	-179
10	72	1360	-256
11	79	1643	-331
12	87	1950	-231
13	94	2304	-380
14	101	2673	-491
15	108	3065	-596
16	115	3481	-699

Table 5: The finite part of Theorem 6.1. The inequality (6.2) covers every $m \geq 17$.

6.2 An open region of dilation failures

Theorem 6.1 is one instance of a general criterion, which identifies an open region of factor data for which the quadratic output must eventually fail. Its proof uses two tools stated later, the first Newton inequality (7.1) of Lemma 7.1 and the asymptotic Lemma 8.1; neither of those depends on anything in this section, so there is no circularity.

Proposition 6.2. *Let $\lambda_1, \dots, \lambda_d$ be positive integers with $d \geq 3$, and let e_2, e_3 be their second and third elementary symmetric functions. For the canonical decomposition of*

$$F_n(t) = \prod_{i=1}^d (1 + n\lambda_i t),$$

the output H_{F_n} is not real-rooted for all sufficiently large integers n , provided

$$9(d-1)^3 e_3^2 > 2(d-2)^3 e_2^3. \quad (6.4)$$

Proof. The input coefficients are $e_2 n^2$ and $e_3 n^3$, so by Lemma 8.1 the first two nonconstant coefficients of H_{F_n} satisfy

$$b_1 = \sqrt{2e_2} n + O(1), \quad b_2 = \frac{(6e_3)^{2/3}}{2} n^2 + O(n).$$

Since H_{F_n} has degree $d-1$, its first Newton difference (Lemma 7.1 with n there equal to $d-1$) is

$$2(d-1)b_2 - (d-2)b_1^2 = ((d-1)(6e_3)^{2/3} - 2(d-2)e_2)n^2 + O(n).$$

The coefficient of n^2 is positive exactly when (6.4) holds, as is seen by cubing the two positive quantities $(d-1)(6e_3)^{2/3}$ and $2(d-2)e_2$. For sufficiently large n the difference is then positive, contradicting (7.1). $\square$

For a cubic $1 + At + Bt^2 + Ct^3$ with $d = 3$, so $e_2 = B$ and $e_3 = C$, criterion (6.4) reduces to

$$B^3 < 36C^2. \quad (6.5)$$

For the simple-root witness (1.4), $B^3 = 26^3 = 17576$ while $36C^2 = 36 \cdot 24^2 = 20736$, so every integer dilation of that single input eventually yields a counterexample for the second output; Theorem 6.1 makes the exceptional set exact in that case. The criterion is strict, hence stable under sufficiently small perturbations of the positive factor parameters. Equality in (6.4) is not decided by this leading-order argument.

7 Uniform constructions in every degree

We now give explicit infinite families in every possible degree. Only the first three coefficients of the second output are needed.

7.1 The first Newton inequality

Lemma 7.1 (First Newton inequality). *If*

$$P(t) = 1 + b_1t + b_2t^2 + \cdots + b_nt^n$$

has positive coefficients, degree $n \geq 2$, and only real zeros, then

$$(n-1)b_1^2 \geq 2nb_2. \quad (7.1)$$

Proof. All roots are negative, so $P = \prod_{i=1}^n (1 + \lambda_i t)$ with $\lambda_i > 0$. By Cauchy–Schwarz,

$$\sum_i \lambda_i^2 \geq \frac{1}{n} \left(\sum_i \lambda_i \right)^2.$$

Since $b_1 = \sum_i \lambda_i$ and $2b_2 = b_1^2 - \sum_i \lambda_i^2$, inequality (7.1) follows. $\square$

This is the first of the usual normalized Newton inequalities; the complete classical sequence is discussed, for example, in [2]. The self-contained proof above is all that is needed here.

Remark 7.2 (The implication runs one way). Failure of (7.1) is sufficient to prove that a polynomial is not real-rooted; satisfaction of (7.1) is *not* sufficient to prove that it is. Every use below is in the first direction. This also limits how the finite Newton certificates of Section 7.4 may be read: a positive margin is a complete proof of failure, whereas a nonpositive margin proves nothing at all.

7.2 The repeated-slope family

Theorem 7.3. *For every integer $d \geq 3$ and every integer $m \geq 20(d-1)$, the canonical second output H for*

$$F(t) = (1 + mt)^d$$

is not real-rooted. In particular, the explicit polynomial

$$(1 + 20(d-1)t)^d$$

is a counterexample in every degree $d \geq 3$.

Proof. Let

$$u = \kappa_2 \left(\binom{d}{2} m^2 \right), \quad v = \kappa_3 \left(\binom{d}{3} m^3 \right).$$

The polynomial H has degree $d-1$, positive coefficients, and initial terms $1 + ut + vt^2$. Its necessary Newton inequality is

$$N := (d-2)u^2 - 2(d-1)v \geq 0.$$

We will prove the strict reverse inequality. Write

$$s = d-1 \geq 2, \quad a = \sqrt{d(d-1)}, \quad \rho = (d(d-1)(d-2))^{1/3} = (s^3 - s)^{1/3}.$$

The two elementary bounds give

$$u < am + 1, \quad v > \frac{\rho^2 m^2}{2} - \frac{3\rho m}{2} + 1.$$

Therefore

$$N < -Am^2 + Bm - d, \tag{7.2}$$

where

$$A = s(\rho^2 - d(d-2)), \quad B = 2(d-2)a + 3s\rho.$$

We establish the elementary estimates $A > s/4$ and $B < 5s^2$. For the first, compare positive cubes:

$$\begin{aligned} (s^3 - s)^2 - \left(s^2 - \frac{3}{4}\right)^3 &= \frac{16s^4 - 44s^2 + 27}{64} \\ &= \frac{16(s^2 - 4)^2 + 84(s^2 - 4) + 107}{64} > 0. \end{aligned}$$

Thus $\rho^2 > s^2 - 3/4$. Since $d(d-2) = s^2 - 1$, this gives $\rho^2 - d(d-2) > 1/4$, as required. For the second estimate, use $a < d$ and $\rho < s$ to obtain

$$B < 2d(d-2) + 3s^2 = 5s^2 - 2 < 5s^2.$$

Substituting in (7.2) yields

$$N < -\frac{s}{4}m^2 + 5s^2m - d = sm \left(5s - \frac{m}{4} \right) - d. \tag{7.3}$$

For $m \geq 20s$, the right-hand side is at most $-d < 0$. This contradicts Lemma 7.1, completing the proof. $\square$

The constant 20 is not optimized. It is useful because it gives a short uniform certificate, without numerical evaluation of radicals or a parameter-dependent unproved threshold. The exact cubic result of Theorem 5.1 is much stronger in degree three.

7.3 The consecutive-slope family with distinct roots

The inputs of Theorem 7.3 have a single repeated root. The next theorem supplies an explicit threshold for a family with d *distinct* roots. Neither theorem implies the other: Theorem 7.3 has the far smaller threshold, and Theorem 7.4 is the only explicit threshold proved here for simple-root inputs. This is pair P3., and both proofs are given in full.

Theorem 7.4. *For integers $d \geq 3$ and $n \geq 10d^3$, set*

$$F_{d,n}(t) = \prod_{j=0}^{d-1} (1 + (n+j)t).$$

Then $F_{d,n}$ has positive integer coefficients and d distinct negative zeros, but its canonical second output $H_{d,n}$ is not real-rooted.

Proof. The input assertions follow from the displayed factorization: each factor parameter $n+j$ is a positive integer, and the d zeros $-1/(n+j)$ are distinct. Put

$$Q = \sqrt{d(d-1)}, \quad R = (d(d-1)(d-2))^{1/3}, \quad K = \frac{d-1}{d-2},$$

and write $H_{d,n}(t) = 1 + b_1 t + b_2 t^2 + \cdots + b_{d-1} t^{d-1}$, a polynomial of degree exactly $d-1$ with positive coefficients.

Every product of two factor parameters is at most $(n+d-1)^2$, and every product of three is at least n^3 , so the input coefficients c_2, c_3 satisfy

$$c_2 \leq \binom{d}{2} (n+d-1)^2, \quad c_3 \geq \binom{d}{3} n^3.$$

By Lemma 2.3, $b_1 = \kappa_2(c_2) < \sqrt{2c_2} + 1 \leq Q(n+d-1) + 1$. By Lemma 2.4 with $x = (6c_3)^{1/3} \geq Rn$, and by monotonicity of $(y-1)(y-2)$ for $y > 3/2$,

$$b_1 < Q(n+d-1) + 1, \quad 2b_2 > (Rn-1)(Rn-2). \quad (7.4)$$

It therefore suffices to prove

$$K(Rn-1)(Rn-2) - (Q(n+d-1) + 1)^2 > 0, \quad (7.5)$$

since that implies $2Kb_2 > b_1^2$, which is the strict reverse of (7.1) for a polynomial of degree $d-1$.

Expanding the left-hand side of (7.5) gives

$$\begin{aligned} \Delta &= KR^2 - Q^2, \\ \Delta n^2 - B_d n + C_d, \quad B_d &= 3KR + 2Q^2(d-1) + 2Q, \\ C_d &= 2K - (Q(d-1) + 1)^2. \end{aligned} \quad (7.6)$$

We use the three uniform bounds

$$\Delta > \frac{1}{3}, \quad B_d < 3d^3, \quad C_d > -d^4. \quad (7.7)$$

For the first, write $u = Q^2 = d(d-1)$ and $v = KR^2$. Direct calculation gives

$$v^3 = \frac{d^2(d-1)^5}{d-2}, \quad u^3 = d^3(d-1)^3, \quad v^3 - u^3 = \frac{d^2(d-1)^3}{d-2} = Ku^2.$$

Hence

$$v^3 - \left(u + \frac{1}{3}\right)^3 = Ku^2 - u^2 - \frac{u}{3} - \frac{1}{27} = \frac{u^2}{d-2} - \frac{u}{3} - \frac{1}{27}.$$

Since $u/(d-2) = d(d-1)/(d-2) > 1$, the right-hand side exceeds $u - u/3 - 1/27 = 2u/3 - 1/27$, and $u \geq 6$ makes this positive. Both v and $u + 1/3$ are positive, so taking cube roots gives $v > u + 1/3$, which is $\Delta > 1/3$.

For the second bound, $Q < d$, $R < d$ and $K \leq 2$ give

$$B_d < 6d + 2d(d-1)^2 + 2d < 2d^3 + 8d \leq 3d^3 \quad (d \geq 3).$$

For the third, $Q(d-1) + 1 < d(d-1) + 1 \leq d^2$ and $K > 0$ give $C_d > -d^4$. Thus (7.6) exceeds

$$\frac{n^2}{3} - 3d^3n - d^4 = n \left(\frac{n}{3} - 3d^3 \right) - d^4.$$

At $n \geq 10d^3$ this is at least $\frac{10}{3}d^6 - d^4 > 0$, and the expression is increasing in n beyond that point. This proves (7.5), so Lemma 7.1 excludes real-rootedness of $H_{d,n}$. $\square$

The number $10d^3$ is an explicit sufficient threshold, not an optimized one, and it is not a claim that $n = 10d^3$ is the first failure. The limiting ratio computed in Section 8.3 approaches one as d increases, while the finite binomial digits introduce rounding effects; a simple bound dominating all of those effects is useful even when far from sharp.

7.4 A compact finite certificate, and how far the bounds are from sharp

For specified d and n , the finite certificate is just the integer

$$M_{d,n} = 2(d-1)\kappa_3(c_3) - (d-2)\kappa_2(c_2)^2, \quad (7.8)$$

where c_2, c_3 are the quadratic and cubic input coefficients. A positive value proves failure, by Lemma 7.1 and Remark 7.2. Neither the expanded input polynomial nor any zero of any output is needed. For example, $d = 3$ and $n = 270$ give

$$c_2 = 220322, \quad c_3 = 19902240, \quad b_1 = 664, \quad b_2 = 121112, \quad M_{3,270} = 43552.$$

The archive records 294 such checks, at $n \in \{10d^3, 10d^3 + 1, 20d^3\}$ for every $3 \leq d \leq 100$, in `results/all_degree_certificates.csv`. That file audits the proved threshold of Theorem 7.4 exactly at and just above its stated value.

Table 6 measures something different, and both are kept; this is pair P11.. It records the first successful *dyadic test values* $m = 1, 2, 4, 8, \dots$ for the repeated-slope inputs $(1 + mt)^d$, with last column the integer N of Theorem 7.3. Each negative entry alone is a complete certificate of failure for the stated pair (d, m) . Already $d = 3$ fails at $m = 8$, far below the proved threshold $20 \cdot 2 = 40$, so the table shows how far from sharp the proved bound is. The certificate file, by contrast, says nothing about sharpness and everything about the correctness of the stated threshold.

d	m	u	v	$(d-2)u^2 - 2(d-1)v$
3	8	20	102	-8
4	16	55	1053	-268
5	32	143	7816	-1181
6	32	175	12390	-1400
7	32	207	18042	-2259
8	32	239	24683	-2836
9	64	543	129574	-9241
10	64	607	164330	-10348
11	64	671	203157	-10971
12	64	735	246217	-14524
13	64	799	293353	-18061
14	64	863	344530	-20552
15	64	927	399787	-22759
16	64	991	459130	-24766
17	128	2111	2091165	-72465
18	128	2239	2361395	-77494
19	128	2367	2647973	-81315
20	128	2495	2950913	-84244

Table 6: Exact Newton-inequality certificates in degrees 3 through 20. The input is $(1 + mt)^d$, and m is the first dyadic test value that works, not necessarily the smallest possible one.

8 Scaling limits and the mechanism of failure

The preceding proofs give explicit counterexamples. A limiting description explains why failure persists and why the two outputs can behave asymmetrically.

8.1 Asymptotics of the integer splitting operator

Lemma 8.1. *Fix an integer $k \geq 2$ and a real number $c > 0$. Suppose q_m are positive integers with*

$$q_m = cm^k + O(m^{k-1}) \quad (m \rightarrow \infty).$$

Then

$$\kappa_k(q_m) = \frac{(k!c)^{(k-1)/k}}{(k-1)!} m^{k-1} + O(m^{k-2}). \quad (8.1)$$

The implicit constant may depend on k , c , and the constant in the hypothesis, but not on m .

Proof. Let $a = a(m)$ be the largest upper index in the k th binomial expansion of q_m . Then

$$\binom{a}{k} \leq q_m < \binom{a+1}{k}.$$

For fixed k , both bounding binomial coefficients equal $a^k/k! + O(a^{k-1})$ as $a \rightarrow \infty$. First this shows $a = \Theta(m)$, and then it shows

$$a^k = k!c m^k + O(m^{k-1}).$$

Writing $\alpha = (k!c)^{1/k}$ and factoring $a^k - (\alpha m)^k$ gives $a = \alpha m + O(1)$.

The largest term of $\kappa_k(q_m)$ is

$$\binom{a-1}{k-1} = \frac{\alpha^{k-1}}{(k-1)!} m^{k-1} + O(m^{k-2}).$$

Every other term has the form $\binom{a_\ell-1}{\ell-1}$ with $\ell \leq k-1$ and $a_\ell < a = O(m)$. There are at most $k-1$ such terms, and each is $O(m^{k-2})$. Their sum can therefore change only the error term, proving (8.1). $\square$

The main term depends only on the leading growth of the input coefficient. The smaller binomial summands, including the jumps at binomial thresholds, are one order lower after rescaling. Equivalently, in terms of the argument itself,

$$\kappa_k(N) = \frac{(k!N)^{(k-1)/k}}{(k-1)!} + O_k(N^{(k-2)/k}) \quad (N \rightarrow \infty),$$

which for $k=2$ is consistent with the exact formula (2.5).

Theorem 8.2. *Fix a polynomial $F(t) = 1 + \sum_{k=1}^d c_k t^k$ with positive integer coefficients, and let $F_m(t) = F(mt)$. Define*

$$Q_F(z) = \sum_{k=1}^d \frac{(k!c_k)^{(k-1)/k}}{(k-1)!} z^{k-1}. \quad (8.2)$$

Then, coefficientwise and uniformly on every compact subset of $\mathbb{C}$,

$$H_{F_m}(z/m) = Q_F(z) + O(m^{-1}), \quad (8.3)$$

$$G_{F_m}(z/m) = F(z) - \frac{z}{m} Q_F(z) + O(m^{-2}). \quad (8.4)$$

Proof. For $k \geq 2$, apply Lemma 8.1 with $q_m = c_k m^k$, and divide by m^{k-1} to obtain the coefficient of z^{k-1} in $H_{F_m}(z/m)$. The constant coefficient, coming from $k=1$, is exactly one, also agreeing with (8.2). There are finitely many coefficients, so the coefficientwise errors are uniform on any prescribed compact set. The identity

$$G_{F_m}(z/m) = F(z) - \frac{z}{m} H_{F_m}(z/m)$$

then gives (8.4). $\square$

Both halves of Theorem 8.2 are used, and this is pair P4.: (8.3) produces the failures, (8.4) produces the recovery in Proposition 8.4, and only the two together exhibit the asymmetry between the outputs.

Remark 8.3 (The limit is not a linear operator). The map $F \mapsto Q_F$ is not a linear transformation of coefficient vectors: different coefficients are raised to different fractional powers with different normalizing constants. Hence the preservation properties of differentiation, or of other linear root-preserving operators, cannot simply be transferred to it.

For the repeated cubic, the limiting quadratic is

$$Q_{(1+t)^3}(z) = 1 + \sqrt{6}z + \frac{6^{2/3}}{2}z^2. \quad (8.5)$$

Its discriminant is $6 - 2 \cdot 6^{2/3} < 0$. This predicts eventual failure of H_m ; Theorem 5.1 supplies the exact exceptional set.

8.2 The first output recovers simple real roots under dilation

Proposition 8.4. *Suppose the fixed polynomial F in Theorem 8.2 has d distinct negative real roots. Then G_{F_m} has d distinct negative real roots for all sufficiently large integers m .*

Proof. Choose d disjoint closed intervals on the negative real axis, each containing one root of F in its interior. Since the roots are simple, choose the endpoints so that F is nonzero and has opposite signs at the two endpoints of each interval. Uniform convergence $G_{F_m}(z/m) \rightarrow F(z)$ preserves those endpoint signs for all sufficiently large m . The intermediate value theorem then gives one root of the rescaled polynomial in each interval.

Also its leading coefficient is $c_d + O(m^{-1}) > 0$ for sufficiently large m , so its degree is d . The d disjoint intervals exhaust its roots, which are therefore all distinct, negative, and real. Rescaling back does not change their reality or multiplicities. $\square$

Applied to $F(t) = (1+2t)(1+3t)(1+4t)$, this proves the eventual real-rootedness of $\tilde{G}_m$ asserted in Section 6.1. Meanwhile H_m is nonreal for every $m \geq 5$. Thus small perturbation of the input polynomial itself is a good model for G , but not for H : the latter converges to the different transform Q_F . This asymmetry — under dilation the first output eventually becomes real-rooted while the second never does — is the most interesting qualitative statement here about the G side, and it is available only because both limits in Theorem 8.2 were proved.

8.3 Simple-root failures in every degree

For $F(t) = (1+t)^d$, write $Q_d = Q_F$. Its first three coefficients are

$$Q_d(z) = 1 + A_1 z + A_2 z^2 + \cdots, \quad A_1 = \sqrt{d(d-1)}, \quad A_2 = \frac{[d(d-1)(d-2)]^{2/3}}{2}. \quad (8.6)$$

Its degree is $d-1$. The first Newton inequality would require

$$d(d-1) \geq \frac{d-1}{d-2} [d(d-1)(d-2)]^{2/3}. \quad (8.7)$$

Both sides are positive, but the ratio of their cubes is

$$\frac{d(d-2)}{(d-1)^2} = 1 - \frac{1}{(d-1)^2} < 1. \quad (8.8)$$

Therefore (8.7) fails strictly for every $d \geq 3$: the limit object Q_d itself violates the first Newton inequality. This gives a qualitative limiting explanation of Theorem 7.3, separate from its explicit bound.

The same limit yields inputs without repeated roots, for which Theorem 7.4 has already supplied an explicit threshold. We record the limiting statement because it is the conceptual reason, and because it applies to consecutive slopes shifted by any fixed amount.

Corollary 8.5. *Fix any integer $d \geq 3$. For all sufficiently large positive integers m , the polynomial*

$$P_{d,m}(t) = \prod_{j=1}^d (1 + (m+j)t) \quad (8.9)$$

has positive integer coefficients and distinct negative roots, while its canonical second output is not real-rooted.

Proof. The input properties are immediate from the distinct positive integer slopes $m+1, \dots, m+d$. For fixed d , its coefficient of t^k is $\binom{d}{k}m^k + O(m^{k-1})$, because it is the sum of $\binom{d}{k}$ products, each with leading term m^k . By Lemma 8.1, the coefficient of z^{k-1} in $H_{P_{d,m}}(z/m)$ tends to the corresponding coefficient of Q_d . The strict reverse of (8.7) is an open inequality in the first two nonconstant coefficients, so it also holds for $H_{P_{d,m}}(z/m)$ once m is sufficiently large. This polynomial has positive coefficients and degree $d-1$, so Lemma 7.1 rules out real-rootedness. Positive rescaling of the variable preserves whether roots are real. $\square$

The phrase “sufficiently large” in Corollary 8.5 is established by a proved coefficient limit and a strict inequality, and no threshold comes out of that argument. The explicit threshold for this family is Theorem 7.4, whose proof is by finite estimates rather than by a limit. Theorem 7.3 has the different explicit threshold $20(d-1)$ for the repeated-slope family, which Theorem 7.4 does not cover.

9 Deletion, link, and a compressed realization

This section reads the identity $F = G + tH$ combinatorially. It explains where a splitting of this shape comes from, and it shows that the failure exhibited in Section 3 is a *local* phenomenon which the f -vector cannot see.

9.1 The deletion–link identity

For a vertex v of a simplicial complex $\mathcal{C}$, the deletion $\text{del}_v \mathcal{C}$ comprises the faces not containing v , and the link $\text{lk}_v \mathcal{C}$ comprises the sets τ disjoint from v for which $\tau \cup \{v\}$ is a face. Separating the faces according to whether they contain v proves

$$f_{\mathcal{C}}(t) = f_{\text{del}_v \mathcal{C}}(t) + t f_{\text{lk}_v \mathcal{C}}(t). \quad (9.1)$$

This is the structural shape of $F = G + tH$. The canonical rule of Definition 2.1 is one arithmetic choice of such a shape, and the next construction realizes that particular choice geometrically for the witness (1.3).

9.2 A compressed complex realizing the full decomposition

Use vertices $1, \dots, 8$. Include the empty face, all eight singletons, every pair from $\{1, \dots, 7\}$ as an edge, and the triangles

$$\binom{[5]}{3} \cup \{\{6\} \cup A : A \in \binom{[4]}{2}\} \cup \{\{1, 5, 6\}, \{2, 5, 6\}\}, \quad (9.2)$$

and no faces of size four or more. All subsets of each displayed triangle are present, so this is a simplicial complex. It has 8 vertices, 21 edges and $10 + 6 + 2 = 18$ triangles, so its face polynomial is exactly the input $1 + 8t + 21t^2 + 18t^3$ of (1.3).

The link of vertex 1 has vertices $2, \dots, 7$; its edges are every pair from $\{2, 3, 4, 5, 6\}$, with vertex 7 isolated. It has no triangular faces, since the complex has no four-vertex faces. Its face polynomial is therefore exactly

$$1 + 6t + \binom{5}{2}t^2 = 1 + 6t + 10t^2 = H(t).$$

The deletion of vertex 1 has 7 vertices, 15 edges and 8 triangles, so its face polynomial is exactly $1 + 7t + 15t^2 + 8t^3 = G(t)$. Direct face counting thus recovers the complete canonical decomposition (1.3), with no arithmetic beyond counting.

The triangles in (9.2) are the first 18 triples in *colexicographic* order, where two distinct sets of the same size are compared by the largest element of their symmetric difference; the edges are the first 21 pairs and the singletons the first 8 elements in the same order. This finite observation explains why a binomially compressed realization appears here. No general compression theorem is invoked for this example.

9.3 The failure is local, not a property of the f -vector

The same input $1 + 8t + 21t^2 + 18t^3$ also has the multipartite realization (3.5) with part sizes 2, 3, 3. Every vertex link in that complex is again a join of discrete sets, so every such link has a real-rooted face polynomial. The compressed complex above has the *same* face-counting polynomial, but its link at vertex 1 is $1 + 6t + 10t^2$, which is not real-rooted. This is the payoff of pair P6.: face counts and their zeros do not determine the real-rootedness of every local link, so the failure discovered in Theorem 3.1 is local and cannot be read off the f -vector.

These two realizations are also different in purpose from the ones in Section 3.3: there the point was that the two *outputs* are themselves f -polynomials, so no realizability obstruction is exhibited on either side. Here the point is that one f -vector supports both a complex all of whose links are real-rooted and a complex with a link that is not. Neither fact follows from the other.

Finally, the coefficientwise conjecture is untouched by either realization: as recorded in Section 3.3, both examples satisfy the coordinatewise inequalities, so Mu–Welker Conjecture 3.8 is not falsified here.

10 Reproducibility and the role of computation

The accompanying files distinguish arithmetic checking from the proofs of infinite statements. Two verification stacks are supplied, and both are kept; this is pair P8.. They are independent in different senses and were run on different software.

The core program `verify.py` requires Python 3.10 or later and only its standard library. It uses arbitrary-precision integers and exact rational arithmetic. There is no numerical root finder, floating-point discriminant, or probabilistic test. The second program, `code/verify_wide.py`, shares no code with it: the canonical expansion, the splitting operator, the discriminants and the search loops are independently reimplemented, and it carries the wider cubic scan, the colex-prefix recomputation and the all-degree certificates.

The counterexamples are hand-checkable proofs. Computation is used to check their arithmetic redundantly, to certify the finite cubic searches in Propositions 4.3 and 4.4, and to generate the finite parts of the exact family classifications. Infinite tails and all-degree conclusions are proved analytically in the text.

10.1 Stack one: the core verifier

Exact check	Scope or count
Two independent greedy implementations compared	21,105 cases
All cubic candidates in (4.2)	621 triples
Retained real-rooted cubic inputs	124 triples
Each of the two cubic families	$1 \leq m \leq 1000$
Printed higher-degree Newton witnesses	$3 \leq d \leq 20$
Additional checks at $m = 20(d - 1)$	$3 \leq d \leq 200$
Independent exact Sturm checks (SymPy 1.14.0)	869 polynomials

The two independent greedy implementations use binary search and linear search, respectively; boundary cases immediately below, at, and above binomial coefficients are included. All checks remain enabled even when Python is run with its optimization flag. Invalid negative, nonintegral, or zero-order inputs are rejected. Output real-rootedness tests remove trailing zero coefficients before deciding the degree. The extended family data in `data/repeated_cubic_family.csv` and `data/simple_cubic_family.csv` cover both cubic families through $m = 1000$ and can be extended with `-family-limit`.

The supplementary `minimal_certificate.py` checks both main examples independently in a short standard-library script, and `independent_sympy_check.py` cross-checks the cubic certificate using exact square-free parts and Sturm root counts on all 621 input polynomials and the 248 outputs of the 124 admissible inputs, recomputing the canonical expansions by a separate linear-search routine. That script does not import `verify.py` and is not a prerequisite for the proofs.

10.2 Stack two: the wide-scan verifier

Exact check	Scope or count
Canonical expansion and Pascal identity, $0 \leq N \leq 5000$, $1 \leq k \leq 8$	40,008 checks
Independent colex-prefix counts	2,509 checks
Real-rooted quadratic inputs, $2 \leq a \leq 100$	84,575 inputs
Candidate cubic triples, $1 \leq a \leq 14$	18,794 triples
All-degree Newton certificates, $3 \leq d \leq 100$	294 certificates
Discriminants recomputed by Sylvester determinant	6 checks

The colex-prefix check is a genuinely different implementation route to the splitting operator: initial colexicographic segments of the k -subsets of $[12]$ are constructed as explicit lists of subsets, and the number of them containing vertex 1 is compared with $\kappa_k(N)$, for each $1 \leq k \leq 6$ and every initial segment. The discriminant check recomputes the same six numbers from the Sylvester determinant of a polynomial and its derivative by exact rational Gaussian elimination, rather than from the closed formula. The explicit complex of Section 9 is machine-enumerated as a face set, checked for closure under subsets and under deletion of a vertex, and checked against the claimed f -, deletion- and link-vectors; the result is recorded in `results/compressed_complex.json` with f -vector $(1, 8, 21, 18)$, deletion vector $(1, 7, 15, 8)$ and link vector $(1, 6, 10)$. The quadratic scan confirms Theorem 4.1 computationally over its stated range, which the proof of course does not require.

The recorded run used Python 3.13.5 and took about 1.28 seconds; that is an observation about that run, not a portable performance claim. Both an ordinary run and a `python -O` run were executed, and the checks raise rather than assert so that optimization does not disable them.

10.3 Two computer algebra systems

The two optional external checks use different systems, and both are recorded. The script `independent_sympy_check.py` uses SymPy 1.14.0, as described above. Separately, `code/verify.wl` was evaluated in a connected Wolfram Language kernel reporting version 15.0.1 for Linux x86 (64-bit); it returned the coefficient lists of all six polynomials of (1.3) and (1.4), both decomposition identities as the zero polynomial, the six discriminants $(4, -31, -3, 0, -59, -4)$ for (F, G, H) of the simple-root example followed by the repeated-root one, real-root counts $(3, 1, 0)$ for the simple-root triple, the symbolic cube difference

$$\frac{(-2 + 3d)(-1 + 8d - 15d^2 + 9d^3)}{27(-2 + d)},$$

and the answer `False` to the question whether any real $d \geq 3$ makes that difference nonpositive — which is the rational positivity step underlying $\Delta > 1/3$ in Theorem 7.4. The exact returned text is in `results/wolfram_output.txt`. Two different systems confirming the same integers is worth more than either alone, and neither is needed to run the standard-library verification or to follow the proofs.

These are independent computational implementations, not independent human refereeing and not a proof-assistant formalization. In particular, the unbounded theorems are not inferred from finite tests.

10.4 Running everything

To regenerate the core certificates and report, run

```
python verify.py
```

from the archive directory; to extend the finite family checks, run

```
python verify.py --family-limit 10000
```

The larger range is optional and does not strengthen an already proved infinite tail. For the second stack,

```
python code/verify_wide.py --out reproduced_results
python code/certificate.py
python minimal_certificate.py
```

Omitting `-out` writes to `results/` and replaces the recorded data there. The files in `data/` contain the complete narrow cubic certificate, extended family data, the higher-degree integer witnesses, and the small generated tables used by this article; the files in `results/` contain the wide cubic scan, the all-degree certificates, the complex certificate and the recorded external output.

To build the PDF after regenerating the tables, run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Two or more ordinary `pdflatex` runs can be used instead. The archive contains the already compiled PDF as well as its source. The bibliography is inline; `references.bib` is supplied only for reference-manager import.

11 Conclusions and precise limits

The proposed preservation principle has a negative answer for the canonical rule in Definition 2.1. The smallest-degree obstruction is already visible in a cubic with eight as its linear coefficient, the smallest with distinct input roots has nine, and in both cases both outputs fail simultaneously. The obstruction persists in infinite families, in every degree at least three, and for inputs with distinct negative roots, with two explicit thresholds proved by different arguments. In high degree, the first Newton inequality alone detects failure; locating nonreal roots numerically is unnecessary.

Several distinctions matter. Both minimal linear coefficients are certified only among cubics. The complete all- m classifications concern G_m, H_m for $(1 + mt)^3$ and the quadratic output for $(1 + 2mt)(1 + 3mt)(1 + 4mt)$, not every possible cubic family. The threshold $20(d - 1)$ applies to the repeated-slope family $(1 + mt)^d$ and the threshold $10d^3$ to the distinct-root consecutive-slope family; neither covers the other, and neither is claimed to be sharp. The all-degree theorems prove failure of H , not simultaneous failure of both outputs in every degree; only the explicit cubic examples prove simultaneous failure. Criterion (6.4) is sufficient, not a classification of every input whose H fails. None of these claims resolves Bell–Skandera’s realization question or the different coefficientwise conjecture of Mu and Welker, and nothing here bears on Proposition 3.9 of [1], which is the bridge between them.

The limiting transform $F \mapsto Q_F$ suggests further concrete problems: determine when this transform is real-rooted, sharpen the uniform thresholds in Theorems 7.3 and 7.4, determine the exact last exceptional value of m in each fixed degree, and decide the equality case of (6.4). These are directions suggested by the present results, not assertions that an exhaustive literature review has verified their novelty or open status.

Finally, mathematical validity and priority are separate issues. The finite counterexamples and the stated proofs can be checked from this manuscript alone. The literature search did not identify a prior answer, but this draft makes no unconditional claim of first discovery. Its bibliographic target is explicitly the versioned question in [1].

A A compact exact search certificate

The following pseudocode in executable Python syntax shows the essential finite enumeration behind Proposition 4.3. Here `decompose` implements Definition 2.1, and `all_real_small` uses the actual polynomial degree and the exact discriminant. Their complete implementations, validation, and tests are supplied in `verify.py`.

```
def D(a, b, c):
    return (a*a*b*b - 4*b*b*b - 4*a*a*a*c
            - 27*c*c + 18*a*b*c)
```

```

failures = []
for a in range(3, 9):
    for b in range(3, a*a // 3 + 1):
        for c in range(1, a*a*a // 27 + 1):
            if D(a, b, c) < 0:
                continue
            g, h = decompose([1, a, b, c])
            if not (all_real_small(g) and all_real_small(h)):
                failures.append((a, b, c))

assert failures == [(8, 21, 18)]

```

The mathematical bounds (4.2) are what make this a complete certificate rather than a search for suggestive examples. Repeated roots are retained by the non-strict input condition $D \geq 0$. Lower-degree outputs are tested separately. Widening the loop to `range(1, 15)` with `b` from 1 gives the scan of Table 3; that variant is implemented in `code/verify_wide.py`.

B A standalone minimal certificate

This is the other half of pair P9.. The program below contains an entire finite disproof by itself, driven from the literal digit table of Table 1. It uses only small binomial identities, explicit coefficient vectors, and the two discriminant formulas, and it depends on no search program and no third-party package. It raises rather than asserts, so it survives `python -O`.

```

from math import comb

def check(condition):
    if not condition:
        raise RuntimeError("Certificate check failed")

digits = [[(9, 1)], [(7, 2), (5, 1)],
           [(6, 3), (3, 2), (1, 1)]]
f, g, h = [1], [1], []
for expansion in digits:
    f.append(sum(comb(a, k) for a, k in expansion))
    g.append(sum(comb(a-1, k) if a-1 >= k else 0
                 for a, k in expansion))
    h.append(sum(comb(a-1, k-1) for a, k in expansion))
check(f == [1, 9, 26, 24])
check(g == [1, 8, 19, 11] and h == [1, 7, 13])
check(all(f[k] == g[k] + h[k-1] for k in range(1, 4)))
check(7**2 - 4*13 == -3)
check(19**2*8**2 - 4*11*8**3 - 4*19**3
      - 27*11**2 + 18*11*19*8 == -31)
print("Exact counterexample verified")

```

The factorization $(1+2t)(1+3t)(1+4t)$ certifies the real zeros of the input; the source definition and the digit table certify that these are the required outputs, rather than merely some decomposition. The file is `code/certificate.py`; the companion `minimal_certificate.py` does the same for *both* witnesses starting from their factorizations instead of their digit tables.

C Source and version audit

The primary source used for the question is the PDF of arXiv:2503.24076v1, dated 31 March 2025. The splitting is on printed page 5, Section 3.2; the target question is on printed page 6 and is numbered 3.10. The PDF page image was checked because the available HTML rendering did not preserve all displayed theorem numbers correctly. The separate coefficientwise conjecture is numbered 3.8 and is also on page 6; the realization question attributed to Bell and Skandera is numbered 1.1; Proposition 3.9 relates 3.8 to 1.1. The source writes f_{i-1} for the coefficient of t^i ; this manuscript writes c_i and keeps the corresponding output indexing, as stated in Section 2.2.

A publisher listing was found for the same work in *Advances in Applied Mathematics*, volume 175 (2026), article 103041, 16 pages, with DOI 10.1016/j.aam.2026.103041. The final publisher full text was not available for inspection during this work; the DOI open attempt returned an access error. It is therefore not assumed that the final article retains the same numbering or that it contains no change to the question. Searches using the title, arXiv identifier, the arXiv DOI 10.48550/arXiv.2503.24076, author names, “Question 3.10”, and “recursive decomposition” with “real rooted” and “counterexample” did not locate an earlier resolution; an author’s publication list at the University of Marburg was also consulted. These searches are evidence of a reasonable check, not proof of worldwide absence of prior work. In particular, inaccessible journal content, unpublished communications, differently worded results, or unindexed corrections could contain an earlier answer.

The full source URLs, search scope, and access limitations are recorded in `REFERENCES.md` and `STATUS.md`; both are retained because neither contains all of the above. No third-party article PDF is redistributed in the archive. This manuscript restates the necessary definitions and presents its counterexamples and proofs directly. No checksum manifest is distributed with this package.

References

- [1] Lili Mu and Volkmar Welker, *On a question about real rooted polynomials and f -polynomials of simplicial complexes*, arXiv:2503.24076v1, 31 March 2025. arXiv:2503.24076v1. Question 3.10 and Section 3.2, on pp. 5–6, are the version-specific target of this manuscript. A publisher listing identifies the journal version as *Advances in Applied Mathematics* **175** (2026), 103041, doi:10.1016/j.aam.2026.103041.
- [2] Jan Vondrák (instructor), Scott Mutchlik (original scribe), *Lecture 11: Real-rooted Polynomials*, Math 233: Non-constructive methods in combinatorics, Stanford University, 2018. Lecture notes on real-rooted polynomials and Newton inequalities. Theorem 11.2 there states Newton’s inequalities; the specific inequality used in this article is proved from scratch in Lemma 7.1, so nothing is used as a black box.